

Practical

1.LINUX

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano linux.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc linux.c -o linux
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./linux
Enter Linux command: date
Parent PID : 611
Child PID : 616
Wed Aug  5 08:34:41 UTC 2026
Child process completed.
sai_pranavi@LAPTOP-RVOHSPKQ:~$ uname
Linux
sai_pranavi@LAPTOP-RVOHSPKQ:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:         39 bits physical, 48 bits virtual
Byte Order:            Little Endian
CPU(s):                12
On-line CPU(s) list:   0-11
Vendor ID:             GenuineIntel
Model name:            13th Gen Intel(R) Core(TM) i5-1335U
CPU family:            6
Model:                 186
Thread(s) per core:    2
Core(s) per socket:    6
Socket(s):             1
Stepping:              3
BogoMIPS:              4992.02
Flags:                 fpu vme de pse tsc msr pae mce cx8 apic sep mtr
r pge mca cmov pat pse36 clflush mmx fxsr sse s
se2 ss ht syscall nx pdpe1gb rdtscp lm constant
_tsc rep_good nopl xtopology tsc_reliable nonst
op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
opcnt tsc_deadline_timer aes xsave avx f16c rdr
and hypervisor lahf_lm abm 3dnowprefetch ssbd i
brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
2 erms invpcid rdseed adx smap clflushopt clwb
sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
vdiri movdir64b fsrm md_clear serialize ibt flu
sh_lld arch_capabilities
```

```
Virtualization features:
Virtualization:        VT-x
Hypervisor vendor:     Microsoft
Virtualization type:   full
Caches (sum of all):
L1d:                   288 KiB (6 instances)
L1i:                   192 KiB (6 instances)
L2:                    7.5 MiB (6 instances)
L3:                   12 MiB (1 instance)
NUMA:
NUMA node(s):          1
NUMA node0 CPU(s):     0-11
Vulnerabilities:
Gather data sampling:   Not affected
Ghostwrite:            Not affected
Indirect target selection: Not affected
Itlb multihit:         Not affected
L1tf:                  Not affected
Mds:                   Not affected
Meltdown:              Not affected
Mmio stale data:       Not affected
Old microcode:         Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed:              Mitigation; Enhanced IBRS
Spec rstack overflow:   Not affected
Spec store bypass:     Mitigation; Speculative Store Bypass disabled v
ia prctl
Spectre v1:            Mitigation; usercopy/swapgs barriers and __user
pointer sanitization
Spectre v2:            Mitigation; Enhanced / Automatic IBRS; IBPB con
ditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_
S
Srbds:                 Not affected
Tsa:                   Not affected
Tsx async abort:       Not affected
Vmscape:               Not affected
sai_pranavi@LAPTOP-RVOHSPKQ:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ps
  PID TTY          TIME CMD
  333 pts/0    00:00:00 bash
  661 pts/0    00:00:00 ps
sai_pranavi@LAPTOP-RVOHSPKQ:~$ top
top - 08:36:48 up 16 min, 1 user, load average: 0.03, 0.03, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7762.6 total,  7135.2 free,   543.2 used,   235.1 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7219.4 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24120	15400	11620	S	0.0	0.2	0:00.61	systemd

```
top - 08:41:54 up 21 min, 1 user, load average: 0.02, 0.04, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7762.6 total,  7126.8 free,   551.6 used,   235.1 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7211.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
671	sai_pra+	20	0	8172	6040	3912	R	0.3	0.1	0:00.18	top
1	root	20	0	24120	15400	11620	S	0.0	0.2	0:00.61	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	50396	16696	15416	S	0.0	0.2	0:00.17	systemd-journal
79	systemd+	20	0	22540	14540	11920	S	0.0	0.2	0:00.07	systemd-resolve
87	root	20	0	35372	12348	9252	S	0.0	0.2	0:00.51	systemd-udev
152	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.05	chronyd-starter
153	root	20	0	4472	3136	2888	S	0.0	0.0	0:00.00	cron
155	message+	20	0	8816	5348	4676	S	0.0	0.1	0:00.05	dbus-daemon
165	root	20	0	44092	29792	14648	S	0.0	0.4	0:00.25	networkd-dispat
172	root	20	0	18432	9400	8156	S	0.0	0.1	0:00.07	systemd-logind
181	syslog	20	0	220548	5788	4508	S	0.0	0.1	0:00.07	rsyslogd
240	root	20	0	123456	32784	18172	S	0.0	0.4	0:00.17	unattended-upgr
248	root	20	0	5492	2712	2532	S	0.0	0.0	0:00.00	agetty
270	_chrony	20	0	24380	11908	7224	S	0.0	0.1	0:00.08	chronyd
272	_chrony	20	0	12092	2452	1808	S	0.0	0.0	0:00.00	chronyd
330	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
331	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.15	Relay(333)
333	sai_pra+	20	0	6268	5520	3852	S	0.0	0.1	0:00.07	bash
334	root	20	0	8644	5564	4724	S	0.0	0.1	0:00.01	login

2. KERNAL SPACE EXECUTION

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano coping.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano input.txt
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano output.txt
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc coping.c -o coping
coping.c: In function 'main':
coping.c:16:50: error: expected expression before '>' token
  16 | while(bytes = read(src, buffer, sizeof(buffer))) > 0)
      |
coping.c:16:53: error: expected statement before ')' token
  16 | while(bytes = read(src, buffer, sizeof(buffer))) > 0)
      |
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano coping.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano coping.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc coping.c -o coping
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./coping
File copied successfully.
sai_pranavi@LAPTOP-RVOHSPKQ:~$
```

3.C PROGRAM USING FORK

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano fork_pid.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc fork_pid.c -o fork_pid
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./fork_pid
Parent Process
PID = 544
Child Process
PID = 545
PPID = 317
```

```
GNU nano 8.7.1 fork_pi
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child Process\n");
        printf("PID = %d\n", getpid());
        printf("PPID = %d\n", getppid());
    }
    else
    {
        printf("Parent Process\n");
        printf("PID = %d\n", getpid());
    }

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano fork_pid.c
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ps -ef
```

```
ps -ef
```

```
top
```

```
cat /proc/$$/status
```

```
PID TTY          PID PPID C   TIME CMD
321 pts/0      00:00:00 bash
574 pts/0      00:00:00 ps

UID PID PPID C   TIME CMD
root 1 0 0 10:50 ? 00:00:01 /sbin/init
root 2 1 0 10:50 hvcb 00:00:00 /init
root 7 2 0 10:50 hvcb 00:00:00 plan9 --control-socket 7
root 47 1 0 10:50 ? 00:00:00 /usr/lib/systemd/systemd
systemd+ 79 1 0 10:50 ? 00:00:00 /usr/lib/systemd/systemd
root 89 1 0 10:50 ? 00:00:00 /usr/lib/systemd/systemd
root 108 1 0 10:50 ? 00:00:00 /bin/sh /usr/lib/systemd
root 112 1 0 10:50 ? 00:00:00 /usr/sbin/cron -f -p
message+ 123 1 0 10:50 ? 00:00:00 @dbus-daemon --system --
root 156 1 0 10:50 ? 00:00:00 /usr/bin/python3 /usr/bi
root 164 1 0 10:50 ? 00:00:00 /usr/lib/systemd/systemd
root 172 1 0 10:50 ? 00:00:00 /usr/libexec/ssl-pro-se
syslog 185 1 0 10:50 ? 00:00:00 /usr/sbin/rsyslogd -n -i
_chrony 258 108 0 10:50 ? 00:00:00 /usr/sbin/chronyd -n -F
_chrony 268 258 0 10:50 ? 00:00:00 /usr/sbin/chronyd -n -F
root 277 1 0 10:50 ? tttyl 00:00:00 /usr/sbin/agetty --noex
root 287 1 0 10:50 ? 00:00:00 /usr/bin/python3 /usr/sh
root 314 2 0 10:50 ? 00:00:00 /init
root 317 314 0 10:50 ? 00:00:00 /init
sai_pra+ 321 317 0 10:50 pts/0 00:00:00 -bash
root 324 2 0 10:50 ? 00:00:00 login -- sai_pranavi
sai_pra+ 384 1 0 10:50 ? 00:00:00 /usr/lib/systemd/systemd
sai_pra+ 388 384 0 10:50 ? 00:00:00 (sd-pam)
sai_pra+ 414 324 0 10:50 pts/1 00:00:00 -bash
sai_pra+ 588 321 0 10:52 pts/0 00:00:00 ps -ef
top - 10:52:44 up 2 min, 1 user, load average: 0.17, 0.21, 0.09
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.
MiB Mem : 7762.6 total, 6622.8 free, 535.6 used, 755.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7227.1 avail Mem
```

```
sai_pra+ 321 0 10:52 pts/0 00:00:00 ps -ef
top - 10:53:08 up 3 min, 1 user, load average: 0.19, 0.21, 0.09
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si,
MiB Mem : 7762.6 total, 6611.1 free, 547.4 used, 755.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7215.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	24340	15364	11552	S	0.0	0.2	0:01.02
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.01
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00
47	root	19	-1	42228	16552	15292	S	0.0	0.2	0:00.17
79	systemd+	20	0	22540	14624	12004	S	0.0	0.2	0:00.09
89	root	20	0	35368	12344	9248	S	0.0	0.2	0:00.30
108	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.04
112	root	20	0	4472	3116	2868	S	0.0	0.0	0:00.01
123	message+	20	0	8812	5388	4732	S	0.0	0.1	0:00.08
156	root	20	0	44092	29704	14560	S	0.0	0.4	0:00.28
164	root	20	0	18428	9408	8168	S	0.0	0.1	0:00.27
173	root	20	0	1792300	15216	12552	S	0.0	0.2	0:00.16
185	syslog	20	0	220548	5932	4624	S	0.0	0.1	0:00.11
258	_chrony	20	0	24116	11672	7144	S	0.0	0.1	0:00.08
268	_chrony	20	0	12092	2444	1800	S	0.0	0.0	0:00.01
277	root	20	0	5492	2736	2548	S	0.0	0.0	0:00.01
287	root	20	0	123456	32592	17980	S	0.0	0.4	0:00.26
314	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00
317	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.01
321	sai_pra+	20	0	6268	5596	3928	S	0.0	0.1	0:00.09
324	root	20	0	8648	5508	4664	S	0.0	0.1	0:00.01
384	sai_pra+	20	0	22336	13524	10904	S	0.0	0.2	0:00.15
388	sai_pra+	20	0	22908	4124	2124	S	0.0	0.1	0:00.00
414	sai_pra+	20	0	6136	5484	3928	S	0.0	0.1	0:00.05
586	sai_pra+	20	0	8172	6012	3904	R	0.0	0.1	0:00.01

4. PARENT PROCESS CREATES MULTIPLE CHILD

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano multi_child.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc multi_child.c -o multi_child
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./multi_child
Child 2 PID=719
Child 1 PID=718
Child 3 PID=720
All Children Completed
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano multi_child.c
GNU nano 8.7.1 multi_child.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    for(int i=1;i<=3;i++)
    {
        if(fork()==0)
        {
            printf("Child %d PID=%d\n",i,getpid());
            return 0;
        }
    }

    while(wait(NULL)>0);

    printf("All Children Completed\n");

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano zombie.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc zombie.c -o zombie
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./zombie.c
-bash: ./zombie.c: Permission denied
4B. sai_pranavi@LAPTOP-RVOHSPKQ:~$
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./zombie
child Exiting
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano zombie.c
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
int main()
{
    if (fork()==0)
    {
        printf("child Exiting\n");
    }
    else {
        sleep(20);
        printf("Parent Finished\n");
    }
    return 0;
}
```

5A. Producer-Consumer using Anonymous Pipe

```
-bash: ./zombie.c: Permission denied
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano zombie.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano pipe_demo.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc pipe_demo.c -o pipe_demo
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./pipe_demo.c
-bash: ./pipe_demo.c: Permission denied
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./pipe_demo
child received: Hello Child
```

```
GNU nano 8.7.1 pipe_demo.c
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>
int main()
{
    int fd[2];
    char msg[] = "Hello Child";
    char buffer[100];
    pipe(fd);
    if(fork()==0)
    {
        close(fd[1]);
        read(fd[0],buffer,sizeof(buffer));
        printf("child received: %s\n", buffer);
        close(fd[0]);
    }
    else {
        close(fd[0]);
        write(fd[1], msg, strlen(msg) + 1);
        close(fd[1]);
        wait(NULL);
    }
    return 0;
}
```

5B. Implement `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()`

```
GNU nano 8.7.1 pipe_exec.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];
    pipe(fd);

    if(fork() == 0)
    {
        close(fd[0]);
        dup2(fd[1], STDOUT_FILENO);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL);
    }

    if(fork() == 0)
    {
        close(fd[1]);
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        execlp("grep", "grep", ".c", NULL);
    }

    close(fd[0]);
    close(fd[1]);

    wait(NULL);
    wait(NULL);

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano pipe_exec.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc pipe_exec.c -o pipe_exec
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./pipe_exec
-rwxr-xr-x 1 sai_pranavi sai_pranavi 16168 Aug 6 04:16 coping
-rw-r--r-- 1 sai_pranavi sai_pranavi 440 Aug 6 04:15 coping.c
-rw-r--r-- 1 sai_pranavi sai_pranavi 100 Jul 30 04:42 display.c
-rw-r--r-- 1 sai_pranavi sai_pranavi 99 Aug 1 05:40 fork.c
-rw-r--r-- 1 sai_pranavi sai_pranavi 340 Aug 9 10:50 fork_pid.c
-rw-r--r-- 1 sai_pranavi sai_pranavi 78 Jul 29 08:37 hello.c
-rw----- 1 sai_pranavi sai_pranavi 63 Jul 29 10:19 hello.c.save
-rw-r--r-- 1 sai_pranavi sai_pranavi 443 Aug 5 08:33 linux.c
-rwxr-xr-x 1 sai_pranavi sai_pranavi 16128 Aug 9 10:57 multi_child
-rw-r--r-- 1 sai_pranavi sai_pranavi 314 Aug 9 10:56 multi_child.c
-rw----- 1 sai_pranavi sai_pranavi 1 Jul 29 06:46 osp.c.save
-rw-r--r-- 1 sai_pranavi sai_pranavi 169 Aug 1 05:48 parent.c
-rwxr-xr-x 1 sai_pranavi sai_pranavi 16048 Aug 1 05:56 parentchild
-rw-r--r-- 1 sai_pranavi sai_pranavi 212 Aug 1 05:56 parentchild.c
-rw-r--r-- 1 sai_pranavi sai_pranavi 379 Aug 9 11:09 pipe_demo.c
-rwxr-xr-x 1 sai_pranavi sai_pranavi 16216 Aug 9 11:15 pipe_exec
-rw-r--r-- 1 sai_pranavi sai_pranavi 500 Aug 9 11:13 pipe_exec.c
drwxr-xr-x 2 sai_pranavi sai_pranavi 4096 Jul 30 03:39 practically
-rw-r--r-- 1 sai_pranavi sai_pranavi 210 Aug 1 06:03 wait.c
-rw-r--r-- 1 sai_pranavi sai_pranavi 159 Aug 9 11:02 zombie.c
```

6A. Named Pipe (FIFO)

```
GNU nano 8.7.1 server.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/stat.h>

int main() {
    char msg[100];
    mkfifo("myfifo", 0666);

    int fd = open("myfifo", O_RDONLY);
    read(fd, msg, sizeof(msg));

    printf("Message from Client: %s\n", msg);

    close(fd);
    return 0;
}
```

```
GNU nano 8.7.1 client.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    char msg[100];

    printf("Enter Message: ");
    fgets(msg, sizeof(msg), stdin);

    int fd = open("myfifo", O_WRONLY);
    write(fd, msg, sizeof(msg));

    close(fd);
    return 0;
}
```

```
Enter Message: hello server
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./client
Enter Message: hello server
sai_pranavi@LAPTOP-RVOHSPKQ:~$
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano server.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano client.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc server.c -o server
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc client.c -o client
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./server
Message from Client: hello server
```

6B. POSIX Signal Handling

```
sai_pranavi@LAPTOP-RVOHSI x Ubuntu
GNU nano 8.7.1 signal.c
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    printf("Signal %d received\n", sig);
}

int main()
{
    signal(SIGINT, handler);
    signal(SIGTERM, handler);
    signal(SIGUSR1, handler);

    while(1)
        pause();

    return 0;
}

sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano signal.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc signal.c -o signal
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./signal
^CSignal 2 received
^CSignal 2 received
```

7A. Memory Layout (Code, Global, Static, Heap, Stack)

```
quit ./signal
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano signal.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano signal.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano memory_layout.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc memory_layout.c -o memory_layout
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./memory_layout
Code Segment : 0x6339148f31a9
Global Variable : 0x6339148f6010
Static Variable : 0x6339148f6014
Heap Variable : 0x63392d7c0010
Stack Variable : 0x7ffd10bd9dfc
```

```
sai_pranavi@LAPTOP-RVOHSI x Ubuntu
GNU nano 8.7.1 memory_layout
#include <stdio.h>
#include <stdlib.h>

int global = 100;
static int stat = 200;

int main()
{
    int stack = 300;
    int *heap = (int *)malloc(sizeof(int));
    *heap = 400;

    printf("Code Segment : %p\n", main);
    printf("Global Variable : %p\n", &global);
    printf("Static Variable : %p\n", &stat);
    printf("Heap Variable : %p\n", heap);
    printf("Stack Variable : %p\n", &stack);

    free(heap);
    return 0;
}
```

7B. /proc/maps

```
sai_pranavi@LAPTOP-RVOHSI  ×  Ubuntu
GNU nano 8.7.1
#include <stdio.h>

int main()
{
    FILE *fp = fopen("/proc/self/maps", "r");
    char line[256];

    while (fgets(line, sizeof(line), fp))
        printf("%s", line);

    fclose(fp);
    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano memory_layout.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano proc_maps.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc proc_maps.c -o proc_maps
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./proc_maps
621f478b5000-621f478b6000 r--p 00000000 08:30 42028 /home/sai_pranavi/proc_maps
621f478b6000-621f478b7000 r-xp 00001000 08:30 42028 /home/sai_pranavi/proc_maps
621f478b7000-621f478b8000 r--p 00002000 08:30 42028 /home/sai_pranavi/proc_maps
621f478b8000-621f478b9000 r--p 00002000 08:30 42028 /home/sai_pranavi/proc_maps
621f478b9000-621f478ba000 rw-p 00003000 08:30 42028 /home/sai_pranavi/proc_maps
621f47c52000-621f47c73000 rw-p 00000000 00:00 0 [heap]
747bbdc00000-747bbdc28000 r--p 00000000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
747bbdc28000-747bbdc8000 r-xp 00028000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
747bbddc0000-747bbde0e000 r--p 001c0000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
747bbde0e000-747bbde12000 r--p 0020d000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
747bbde12000-747bbde14000 rw-p 00211000 08:30 14176 /usr/lib/x86_64-linux-gnu/libc.so.6
747bbde14000-747bbde21000 rw-p 00000000 00:00 0
747bbe004000-747bbe007000 rw-p 00000000 00:00 0
747bbe00d000-747bbe00f000 rw-p 00000000 00:00 0
747bbe00f000-747bbe013000 r--p 00000000 00:00 0 [vvar]
747bbe013000-747bbe015000 r--p 00000000 00:00 0 [vvar_vclock]
747bbe015000-747bbe017000 r-xp 00000000 00:00 0 [vdso]
747bbe017000-747bbe018000 r--p 00000000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
747bbe018000-747bbe047000 r-xp 00001000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
747bbe047000-747bbe052000 r--p 00030000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
747bbe052000-747bbe054000 r--p 0003b000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
747bbe054000-747bbe055000 rw-p 0003d000 08:30 13931 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
747bbe055000-747bbe056000 rw-p 00000000 00:00 0
7ffd3c466000-7ffd3c487000 rw-p 00000000 00:00 0 [stack]
sai_pranavi@LAPTOP-RVOHSPKQ:~$
```

8A. malloc(), calloc(), realloc(), free()

```
sai_pranavi@LAPTOP-RVOHSI  ×  Ubuntu
GNU nano 8.7.1 memory.c
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *a = (int *)malloc(5 * sizeof(int));

    for(int i=0; i<5; i++)
        a[i]=i+1;

    a = realloc(a, 10*sizeof(int));

    for(int i=5; i<10; i++)
        a[i]=i+1;

    printf("Array:\n");
    for(int i=0; i<10; i++)
        printf("%d ", a[i]);

    free(a);

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano memory.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc memory.c -o memory
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./memory
Array:
1 2 3 4 5 6 7 8 9 10 sai_pranavi@LAPTOP-RVOHSPKQ:~$
```

8B. malloc(), calloc(), realloc(), free()

```
sai_pranavi@LAPTOP-RVOHSI x Ubuntu
GNU nano 8.7.1 cow.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    int x=10;

    if(fork()==0)
    {
        x=20;
        printf("Child x=%d\n",x);
    }
    else
    {
        sleep(1);
        printf("Parent x=%d\n",x);
    }

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc cow.c -o cow
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./cow
Child x=20
Parent x=10
```

9A. File Copy using Low-Level System Calls

```
sai_pranavi@LAPTOP-RVOHSI x Ubuntu
GNU nano 8.7.1 filecopy.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest, n;
    char buffer[1024];

    src = open("source.txt", O_RDONLY);
    dest = open("dest.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while((n = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, n);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano filecopy.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ^C
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc filecopy.c -o filecopy
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./filecopy
File copied successfully.
```

9B. I/O Redirection using dup2()

```
sai_pranavi@LAPTOP-RVOHSI x Ubuntu
GNU nano 8.7.1 dup2_demo.c *
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main()
{
    int fd = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    dup2(fd, STDOUT_FILENO);

    printf("Hello Linux\n");

    close(fd);

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano dup2_demo.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc dup2_demo.c -o dup2_demo
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./dup2_demo
sai_pranavi@LAPTOP-RVOHSPKQ:~$ cat output.txt
Hello Linux
```

10A. Hard Link & Symbolic Link

```
sai_pranavi@LAPTOP-RVOHSI  ×  Ubuntu
GNU nano 8.7.1                inode_demo.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    link("file1.txt", "hardlink.txt");
    symlink("file1.txt", "softlink.txt");

    printf("Hard link and Symbolic link created.\n");

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano inode_demo.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc inode_demo.c -o inode_demo
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./inode_demo
ls -li
Hard link and Symbolic link created.
total 512
53132 -rw-r--r-- 1 sai_pranavi sai_pranavi    37 Jul 29 08:15 1.txt
53158 -rw-r--r-- 1 sai_pranavi sai_pranavi    19 Jul 29 08:21 2.txt
53137 -rw-r--r-- 1 sai_pranavi sai_pranavi     0 Jul 29 08:09 22.txt
53145 -rw-r--r-- 1 sai_pranavi sai_pranavi    21 Jul 29 07:11 3.txt
53150 drwxr-xr-x 2 sai_pranavi sai_pranavi  4096 Jul 29 08:22 4.txt
53217 drwxr-xr-x 2 sai_pranavi sai_pranavi  4096 Jul 30 04:22 OSLAB
```

10B. mmap() File Reading

```
sai_pranavi@LAPTOP-RVOHSI  ×  Ubuntu
GNU nano 8.7.1                mmap_demo.c *
#include <stdio.h>
#include <fcntl.h>
#include <sys/mman.h>
#include <unistd.h>

int main()
{
    int fd = open("sample.txt", O_RDONLY);

    char *data = mmap(NULL, 100, PROT_READ, MAP_PRIVATE, fd, 0);

    printf("%s\n", data);

    munmap(data, 100);
    close(fd);

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano mmap_demo.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc mmap_demo.c -o mmap_demo
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./mmap_demo
Segmentation fault
./mmap_demo
```

11A. POSIX Threads (Race Condition)

```
sai_pranavi@LAPTOP-RVOHSI x Ubuntu x
GNU nano 8.7.1 pthread_demo.c *
#include <stdio.h>
#include <pthread.h>

int counter = 0;

void *increment(void *arg)
{
    for(int i = 0; i < 100000; i++)
        counter++;

    return NULL;
}

int main()
{
    pthread_t t1, t2;

    pthread_create(&t1, NULL, increment, NULL);
    pthread_create(&t2, NULL, increment, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    printf("Counter = %d\n", counter);

    return 0;
}

sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano pthread_demo.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc pthread_demo.c -o pthread_demo -lpthread
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./pthread_demo
Counter = 127042
```

11B. Mutex Lock

```
sai_pranavi@LAPTOP-RVOHSI x Ubuntu x
GNU nano 8.7.1 mutex_demo.c *
#include <stdio.h>
#include <pthread.h>

int counter = 0;
pthread_mutex_t lock;

void *increment(void *arg)
{
    for(int i = 0; i < 100000; i++)
    {
        pthread_mutex_lock(&lock);
        counter++;
        pthread_mutex_unlock(&lock);
    }

    return NULL;
}

int main()
{
    pthread_t t1, t2;

    pthread_mutex_init(&lock, NULL);

    pthread_create(&t1, NULL, increment, NULL);
    pthread_create(&t2, NULL, increment, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    pthread_mutex_destroy(&lock);

    printf("Counter = %d\n", counter);

    return 0;
}

sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano mutex_demo.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc mutex_demo.c -o mutex_demo -lpthread
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./mutex_demo
Counter = 200000
```

12A. Producer-Consumer using Semaphores

```
GNU nano 8.7.1 producer_consumer.c
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>

#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE];
int in = 0, out = 0;

sem_t empty;
sem_t full;
pthread_mutex_t mutex;

void *producer(void *arg)
{
    for (int i = 1; i <= 5; i++)
    {
        sem_wait(&empty);

        pthread_mutex_lock(&mutex);

        buffer[in] = i;
        printf("Produced: %d\n", i);
        in = (in + 1) % BUFFER_SIZE;

        pthread_mutex_unlock(&mutex);

        sem_post(&full);
    }

    return NULL;
}

void *consumer(void *arg)
{
    for (int i = 1; i <= 5; i++)
    {
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano producer_consumer.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc producer_consumer.c -o producer_consumer -pthread
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./producer_consumer
Produced: 1
Produced: 2
Produced: 3
Produced: 4
Produced: 5
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
Consumed: 5
```

12B. Deadlock Prevention (Resource Ordering)

```
sai_pranavi@LAPTOP-RVOHSI  ×  Ubuntu
GNU nano 8.7.1                deadlock_prevention.c *
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>

pthread_mutex_t lock1;
pthread_mutex_t lock2;

void *thread1(void *arg)
{
    pthread_mutex_lock(&lock1);
    printf("Thread 1 acquired Resource 1\n");

    sleep(1);

    pthread_mutex_lock(&lock2);
    printf("Thread 1 acquired Resource 2\n");

    printf("Thread 1 is working\n");

    pthread_mutex_unlock(&lock2);
    pthread_mutex_unlock(&lock1);

    return NULL;
}

void *thread2(void *arg)
{
    /* Resource ordering:
       Always acquire lock1 before lock2 */

    pthread_mutex_lock(&lock1);
    printf("Thread 2 acquired Resource 1\n");

    sleep(1);

    pthread_mutex_lock(&lock2);
    printf("Thread 2 acquired Resource 2\n");
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano producer_consumer.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano deadlock_prevention.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc deadlock_prevention.c -o deadlock_prevention -pthread
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./deadlock_prevention
Thread 2 acquired Resource 1
Thread 2 acquired Resource 2
Thread 2 is working
Thread 1 acquired Resource 1
Thread 1 acquired Resource 2
Thread 1 is working
Deadlock prevented using resource ordering.
```